

FORBES, Australia

WMO: 947150

Lat: 33.3628S

Lon: 147.9206E

Elev: 758

StdP: 14.30

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.2	32.4	20.0	15.5	72.5	23.9	18.6	64.3	23.1	57.1	20.9	56.3	1.6	180	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	101.6	66.5	97.9	65.8	94.6	65.3	72.1	84.0	70.8	83.8	69.5	83.9	11.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	113.6	73.7	67.8	105.2	73.6	65.5	96.8	73.1	36.4	83.3	35.2	83.3	34.0	83.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	20.0	17.7	DB	26.1	107.1	2.5	3.7	24.3	109.7	22.8	111.8	21.4	113.9	19.6	116.6
			WB	25.8	74.7	2.5	1.8	24.1	76.1	22.6	77.1	21.2	78.2	19.4	79.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.3	79.2	77.0	71.0	62.6	54.2	49.0	47.5	49.0	54.6	61.7	68.7	73.8
	DBStd	12.45	6.54	6.34	6.04	5.85	5.25	4.56	4.16	4.43	5.53	6.74	7.25	6.60
	HDD50	281	0	0	0	1	19	72	101	73	14	1	0	0
	HDD65	2499	0	1	17	110	338	479	544	497	317	149	40	7
	CDD50	4757	906	756	651	378	148	43	22	41	151	363	561	737
	CDD65	1499	441	337	203	37	1	0	0	0	4	45	151	280
	CDH74	19290	5810	4019	2141	504	18	0	0	2	115	802	2166	3713
	CDH80	9608	3320	2117	876	118	0	0	0	0	18	278	1010	1871
(14) Wind	WSAvg	7.8	9.6	8.9	8.2	6.6	6.2	6.5	6.6	7.1	7.8	8.2	9.2	9.2
(15) Precipitation	PrecAvg	22.0	2.4	1.7	1.6	1.6	2.1	1.3	1.9	1.9	1.9	2.2	1.9	1.8
	PrecMax	36.8	9.1	6.7	5.0	11.8	6.1	3.1	5.7	4.2	5.8	7.1	6.9	5.4
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.2	0.0	0.2	0.0	0.4	0.2	0.1	0.0
	PrecStd	6.4	2.2	1.7	1.3	2.1	1.5	0.8	1.2	1.1	1.4	1.5	1.6	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.5	104.9	96.4	88.8	76.3	68.2	66.6	72.1	82.4	91.8	101.5	103.1
		MCWB	67.8	68.9	64.5	62.5	59.3	55.6	53.8	54.8	58.1	60.5	64.0	65.2
	2%	DB	102.5	99.6	92.1	83.4	72.7	64.2	62.7	67.1	77.1	87.1	95.4	97.7
		MCWB	66.7	67.2	64.3	61.5	57.8	54.6	52.5	53.7	56.9	60.3	62.6	64.7
	5%	DB	98.8	95.4	88.4	79.7	69.5	61.4	59.7	63.8	73.1	82.6	90.2	93.9
		MCWB	66.8	66.6	64.1	60.1	56.5	53.5	51.9	52.2	56.2	59.3	61.8	63.9
	10%	DB	94.9	91.7	84.9	76.2	65.9	58.7	57.1	60.6	69.0	78.1	85.6	90.1
		MCWB	66.3	65.7	63.4	59.0	55.0	52.9	50.6	51.6	55.2	58.6	60.7	63.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	74.8	72.5	67.2	63.3	59.5	59.2	59.2	63.7	66.7	69.4	72.5
		MCDB	88.0	82.2	79.0	75.8	68.0	61.8	60.9	64.3	71.9	76.5	81.0	82.1
	2%	WB	71.8	72.4	70.1	64.7	60.4	57.0	55.2	56.1	61.0	64.5	67.3	70.0
		MCDB	87.1	83.6	78.4	74.2	68.3	60.3	59.4	62.8	69.8	75.4	80.9	81.5
	5%	WB	70.5	70.9	68.2	62.8	58.4	55.2	53.1	54.2	59.0	62.8	65.7	68.2
		MCDB	87.3	83.2	78.6	73.6	65.7	59.7	57.3	61.5	68.2	74.8	79.2	81.4
	10%	WB	69.2	69.4	66.4	61.1	56.7	53.5	51.2	52.0	56.9	61.0	63.9	66.4
		MCDB	86.9	83.5	78.7	72.7	64.1	57.9	55.8	59.3	65.4	73.7	77.9	82.1
(35) Mean Daily Temperature Range	5% DB	MDBR	26.7	24.9	24.3	25.4	23.1	19.3	19.5	22.7	26.0	27.3	27.4	27.4
		MCDBR	30.9	29.6	29.0	29.9	27.6	23.5	24.6	28.6	33.2	34.2	33.8	31.9
		MCWBR	9.1	9.6	10.4	12.7	14.5	14.6	15.2	16.2	16.1	13.6	11.1	10.2
	5% WB	MCDBR	24.4	21.6	21.0	23.3	21.5	18.7	18.8	23.2	24.9	25.8	24.5	24.0
		MCWBR	9.2	8.6	9.5	11.0	12.2	12.6	12.9	14.5	14.0	12.1	9.4	9.8
(40) Clear-Sky Solar Irradiance	taub		0.346	0.347	0.323	0.306	0.291	0.285	0.281	0.288	0.318	0.328	0.343	0.338
	taud		2.498	2.523	2.568	2.584	2.585	2.608	2.610	2.567	2.467	2.468	2.470	2.508
	Ebn at Noon		314	308	304	293	281	275	282	295	301	310	313	318
	Edh at Noon		36	34	30	27	23	22	23	27	33	36	37	36
(44) All-Sky Solar Radiation	RadAvg		2444	2129	1807	1413	1031	792	888	1182	1641	2029	2245	2464
	RadStd		120	153	118	93	62	65	58	89	120	148	197	100

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	+0.85	N/A	N/A	+2.30	N/A	N/A	N/A	-158	+344	+202

Nomenclature: See separate page